FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE JEE (Advanced), 2015

CONCEPT RECAPITULATION TEST - II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 16) contains 4 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (17 – 20) contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

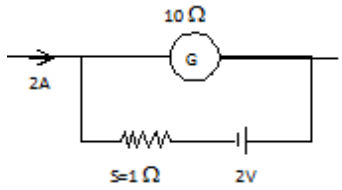
Multi Correct Choice Type

This section contains **8 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. Suppose the potential energy between electron and proton at a distance r is given by $\frac{-ke^2}{3r^3}$. Application of Bohr's theory to hydrogen atom in this case shows that
   (A) Energy in the nth orbit is proportional to n^6
   (B) Energy is proportional to $m^{-3}.(m:\text{mass of electron})$
   (C) Energy of the nth orbit is proportional to n^{-2}
   (D) Energy is proportional to m^3 (m = mass of electron)

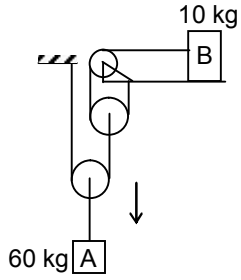
2. During an experiment an ideal gas is found to obey a condition $\frac{P^2}{\rho}$ = constant (ρ = density of gas). The gas is initially at temperature T_0, pressure P_0 and density ρ_0. The gas expands such that density changes to $\frac{\rho_0}{2}$. Then which of the following is/are correct:
   (A) The pressure of the gas changes to $\sqrt{2}P_0$
   (B) The temperature of the gas changes to $\sqrt{2}T_0$
   (C) The graph of the above process on the P – T diagram is parabola.
   (D) The graph of the above process on the P – T diagram is hyperbola.

3. The galvanometer shown in the figure has resistance 10Ω. It is shunted by a series combination of a resistance S = 1Ω and an ideal cell of emf 2V. A current 2A passes as shown. Then choose the correct statement(s).
   (A) The reading of the galvanometer is 1A
   (B) The reading of the galvanometer is zero
   (C) The potential difference across the resistance S is 1.5 V
   (D) The potential difference across the resistance S is 2V



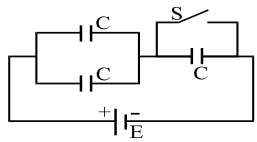
Space for Rough work

4. Figure shows two blocks A and B connected to an ideal pulley string system. In this system when bodies are released then:



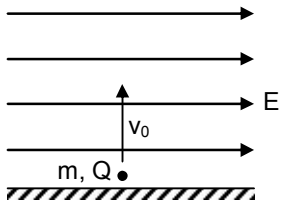
(A) Acceleration of block A is 1 m/s^2
(B) Acceleration of block B is 2 m/s^2
(C) Tension in string connected to block B is 40 N
(D) Tension in string connected to block B is 80 N

5. In the circuit shown, each capacitor has a capacitance C. The emf of the cell is E. If the switch S is closed



(A) positive charge will flow out of the positive terminal of the cell
(B) positive charge will enter the positive terminal of the cell
(C) The amount of charge flowing through the cell will be CE.
(D) The amount of charge flowing through the cell will be 4/3 CE.

6. A point-like body of mass m and of charge Q is initially at rest on the horizontal tabletop as shown in the figure and is given a v_0 vertical, upward speed. There is a horizontal electric field E. Acceleration due to gravity = g.



(A) Maximum height attained $= \frac{v_0^2}{2g}$

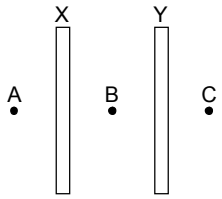
(B) Particle will collide with the table at horizontal distance $\frac{2QEv_0^2}{mg^2}$

(C) At the highest point acceleration will be only in horizontal direction.

(D) Particle will collide with the table at an angle $\tan\theta = \frac{mg}{2QE}$ with horizontal.

Space for Rough work

7. X and Y are large parallel conducting plates close to each other. Each face has an area A. X is given a charge Q, Y is without any charge. Point A, B and C are shown.



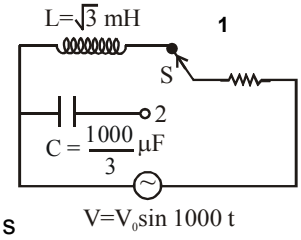
(A) Electric field at B is $\frac{Q}{2\varepsilon_0 A}$

(B) Electric field at B is $\frac{Q}{\varepsilon_0 A}$

(C) The fields at A, B and C are of same magnitude

(D) The fields at A and C are in opposite direction.

8. In the given AC circuit, when switch S is at position 1, the source emf leads current by $\frac{\pi}{6}$. Now, if the switch is at position 2, then



(A) current leads source emf by $\frac{\pi}{4}$

(B) current leads source emf by $\frac{\pi}{3}$

(C) source emf leads current by $\frac{\pi}{4}$

(D) source emf leads current by $\frac{\pi}{3}$

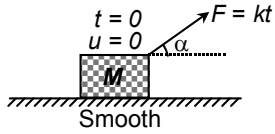
Space for Rough work

Comprehension Type

This section contains **4 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Questions 09 & 10

A block of mass *M* is kept on a smooth horizontal floor. A variable force *F = kt* is acting on the block at an angle of α with the horizontal. At initial moment the block is at rest (the situation shown in the figure below). The block leaves the contact with floor at time $t = t_0$



9. The value of t_0 is

(A) $\frac{Mg}{k\cos\alpha}$ (B) $\frac{Mg}{k\sin\alpha}$

(C) $\frac{Mg\tan\alpha}{k}$ (D) $\frac{Mg\cot\alpha}{k}$

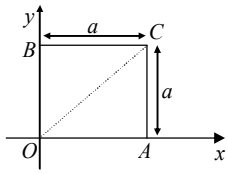
10. The displacement of the block till time $t = t_0$ is

(A) $\frac{M^2g^3\cot\alpha\ \text{cosec}^2\alpha}{6k^2}$ (B) $\frac{M^2g^3\cot\alpha\ \text{cosec}^2\alpha}{2k^2}$

(C) $\frac{M^2g^3\tan\alpha\ \sec^2\alpha}{6k^2}$ (D) $\frac{M^2g^3\tan\alpha\ \sec^2\alpha}{2k^2}$

Paragraph for Questions 11 & 12

A force acting on a particle depends on the particles position in the x-y plane. This force is given by the expression $\vec{F} = x^2y^2\hat{i} + x^2y^2\hat{j}$(N). The particle is moved in the x-y plane along various paths as shown in the figure.



11. Work done by the force along the path *OAC* is

(A) $\frac{a^5}{3}$J (B) $\frac{a^4}{9}$J

(C) $\frac{2a^5}{5}$J (D) zero

12. Work done by the force along the path *OC* is

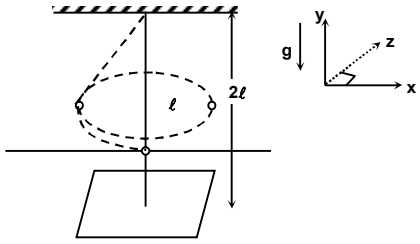
(A) $\frac{a^5}{3}$J (B) $\frac{3a^5}{8}$J

(C) $\frac{2a^5}{5}$J (D) $-\frac{a^5}{3}$J

Space for Rough work

Paragraph for Questions 13 & 14

A pendulum bob has mass m. The length of pendulum is ℓ. It is initially at rest. A particle P of mass m/2 moving horizontally along –ve x-direction with velocity $\sqrt{2g\ell}$ collides with the bob and comes to rest. When the bob comes to rest momentarily, another particle Q of mass m moving horizontally along z-direction collides with the bob and sticks to it. It is observed that the bob now moves along a horizontal circle. The floor is a horizontal surface at a distance 2ℓ below the point of suspension of the pendulum.



13. Tension in string immediately after the first collision is

(A) 2mg (B) mg

(C) $\frac{3}{2}$ mg (D) $\frac{5}{2}$ mg

14. The height of circular path above the floor is

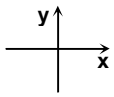
(A) $\frac{3\ell}{2}$ (B) $\frac{4\ell}{3}$

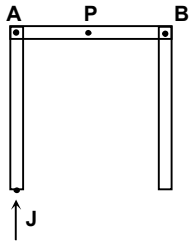
(C) $\frac{5\ell}{4}$ (D) data not sufficient

Space for Rough work

Paragraph for Questions 15 & 16

A uniform wire frame of linear mass density λ having three sides each of length 2a is kept on a smooth horizontal surface. An impulse J is applied at one end as shown in the figure. P is the midpoint of AB. Now answer the following question.





15. The angular velocity of system just after the impulse

(A) $\frac{3J}{22\lambda a^2}$ (B) $\frac{J}{22\lambda a^2}$

(C) $\frac{2J}{22\lambda a^2}$ (D) $\frac{4J}{22\lambda a^2}$

16. The velocity of point P just after the impulse is

(A) $\frac{J}{\lambda a}\hat{j}$ (B) $\frac{J}{6\lambda a}\hat{j}$

(C) $\frac{J}{\lambda a}\left(\frac{2}{11}\hat{i}+\frac{1}{6}\hat{j}\right)$ (D) $\frac{J}{\lambda a}\left(\frac{1}{11}\hat{i}+\frac{1}{6}\hat{j}\right)$

Space for Rough work

(Match list Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

17. In L – C – R series circuit suppose ω_r is the resonance frequency, then match the following table

	List-I		**List-II**
(P)	If $\omega > \omega_r$	1.	Current will lead the voltage
(Q)	If $\omega = \omega_r$	2.	Voltage will lead the current
(R)	If $\omega = 2\omega_r$	3.	$X_L = 2X_C$
(S)	If $\omega < \omega_r$	4.	Current and voltage are in phase

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	2	4	2	1
(C)	1	2	3	4
(D)	2	4	3	1

18. A ray of light falls normally on an equilateral prism of refractive index $\mu = \sqrt{3}$. Match the following table :

i_1 = Angle of incidence
i_2 = Angle of emergence from second surface of prism
r_1 = Angle of refraction at the first surface of prism
r_2 = Angle of incidence at second surface of prism

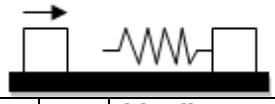
	List-I		**List-II**
(P)	Angle i_1	1.	0°
(Q)	Angle r_1	2.	90°
(R)	Angle i_2	3.	60°
(S)	Angle r_2	4.	None

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	2	3	4	1
(C)	1	2	3	4
(D)	1	1	4	3

Space for Rough work

19. In the arrangement shown in fig. match the following:

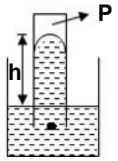


	List-I		List-II
(P)	Velocity of centre of mass	1.	2 SI unit
(Q)	Velocity of combined mass when compression in the spring is maximum	2.	1 SI unit
(R)	Maximum compression in the spring	3.	4 SI unit
(S)	Maximum potential energy stored in the spring	4.	0.5 SI unit

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	2	3	4	1
(C)	2	2	2	1
(D)	1	2	2	1

20. A tube is inverted in a mercury vessel as shown in fig. P is the pressure inside the tube above the mercury level as shown.



	List-I		List-II
(P)	If P is increased, then height h	1.	Increase
(Q)	If P is increased, then pressure at O	2.	Decreases
(R)	If P is increased, then pressure at 1 cm above O	3.	Remains same
(S)	If the vessel is moved upwards with acceleration (a<g), then height h	4.	Increase or decreases depending on the size of the tube

Codes:

	P	Q	R	S
(A)	4	2	1	3
(B)	2	3	3	2
(C)	1	2	3	4
(D)	2	4	3	1

Space for Rough work

Chemistry

PART – II

SECTION – A

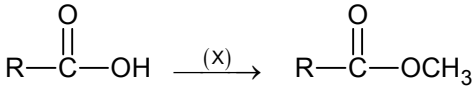
Straight Objective Type

This section contains **8 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE** or **MORE** than is/are correct.

1. Select pair(s) in which both produces same coloured ppt. with CrO_4^{2-} ion
   (A) Ba^{2+}, Sr^{2+} (B) Pb^{2+}, Ba^{2+}
   (C) Ag^{+}, Ba^{2+} (D) Ag^{+}, Hg_2^{2+}

2. Which of the following property (s) is/are related to solution of rosy red complex of Ni^{2+} and dimethyl glyoximate ligand?
   (A) Geometry around Ni is square planar and diamagnetic in nature
   (B) Complex is stabilized by internal hydrogen bonding
   (C) it is an organometallic complex
   (D) Five member chelate ring is formed when each molecule of dimethyl glyoximate is bonded with metal ion.

3. In the given reaction



   X will be
   (A) CH_2N_2 (B) CH_3OH/H^+
   (C) MeCOOH (D) Me_2SO_4

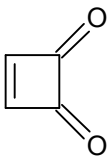
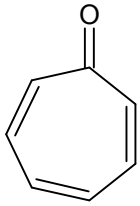
4. In a cell, $Zn(s) \mid Zn^{2+} \mid\mid H^+ \mid H_2$ (pt)
   the addition of H_2SO_4 to the cathode compartment will
   (A) Decrease E (B) Increase E
   (C) Shift equilibrium to left (D) Shift equilibrium to right

Space for Rough work

5. Select the correct staement (s) among the following
   (A) Increase in concentration of reactant increases the rate of zero order reaction
   (B) Rate constant K is equal to collision frequency A, if $E_a = 0$
   (C) Rate constant K is equal to collision frequency A if $E_a = \infty$
   (D) Graph of $\log_{10}K$ vs 1/T is straightline

6. Self reduction is not used in the extraction of
   (A) Lead (B) Manganese
   (C) Copper (D) Iron

7. The Cl_2 evolve at anode during electrolysis of fused $MgCl_2$ which produces 13 g Mg. (Atomic mass of Mg = 24.3) is
   (A) 11.98 lit (B) 0.535 mole
   (C) 37.98 gm (D) 1.068 mole

8. Which are aromatic compounds?
   (A) 
   (B) 
   (C) 
   (D) 

Space for Rough work

Comprehension Type

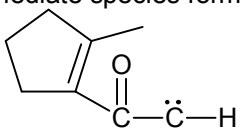
This section contains **4 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

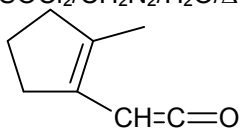
Paragraph for Question Nos. 9 to 10

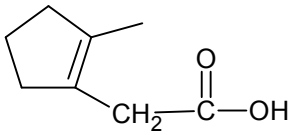
An organic compound (X) is subjected to reductive ozonolysis to give a product (A) which on heating with dil. NaOH gives another product (B). Compound (B) on heating with sodium hypoiodite followed by acidification produces (C). Compound (C) on treatment with $SOCl_2$ and then with diazomethane followed by hydrolysis gives a product which on heating gives 1, 2-dimethyl cyclopentene.
Now answer the following questions.

9. Organic compound (X) in the above passage is:
   (A) 1,2-dimethyl cyclopentene (B) 1,2-dimethyl hexene
   (C) 1,2-dimethyl cyclohexene (D) 2,5-dimethyl-3-hexene.

10. Intermediate species formed in the reaction of (C) with $SOCl_2/CH_2N_2/H_2O/\Delta$ is:

(A) 

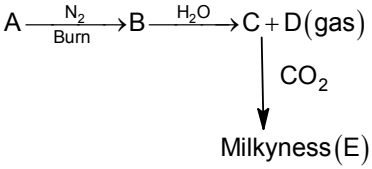
(B) 

(C) 

(D) All of these

Space for Rough work

Paragraph for Question Nos. 11 to 12

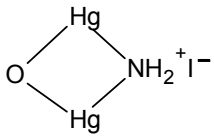


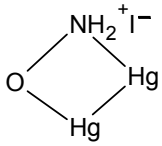
$ACl_X \longrightarrow$ green colour in flame.

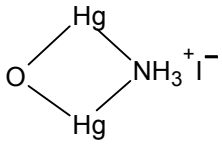
11. What is "A"?
(A) Ca
(B) Mg
(C) Ba
(D) None

12. $KI(\text{excess}) + HgI_2 \longrightarrow F \xrightarrow[KOH]{D} G$

G is

(A) 

(B) 

(C) 

(D) None

Space for Rough work

Paragraph for Question Nos. 13 to 14

Redox reactions play a vital role in chemistry and biology. The values of E^{o}_{red} of two half cell reactions decide which way the reaction is expected to proceed. A simple example is Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E^{o} values.

$I_2 + 2e^- \longrightarrow 2I^-$; $E^o = 0.54V$

$Cl_2 + 2e^- \longrightarrow 2Cl^-$; $E^o = 1.36V$

$Mn^{3+} + e^- \longrightarrow Mn^{2+}$; $E^o = 1.50V$

$Fe^{3+} + e^- \longrightarrow Fe^{2+}$; $E^o = 0.77V$

$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$; $E^o = 1.23V$

13. Fe^{3+} is stable but Mn^{3+} is not stable in acidic solution because:
(A) O_2 oxidises Mn^{2+} to Mn^{3+}
(B) O_2 oxidises both Mn^{2+} and Fe^{2+} into Mn^{3+} and Fe^{3+} respectively.
(C) Fe^{3+} oxidises H_2O to O_2
(D) Mn^{3+} oxidises H_2O to O_2.

14. Among the following identify the correct statement:
(A) Chloride ion is oxidized by O_2
(B) Fe^{2+} ion is oxidized by iodine
(C) Iodide ion is oxidized by chlorine
(D) Mn^{2+} is oxidized by chlorine.

Space for Rough work

Paragraph for Question Nos. 15 to 16

When 20.02 g of a white solid (X) is heated, 4.4 g of an acid gas (A) and 1.8 g of a neutral gas (B) are evolved leaving behind a solid residue (Y) of weight 13.8 g (A) turns lime water milky and (B) condenses into a liquid which changes anhydrous copper sulphate blue. The aqueous solution of Y is alkaline to litmus and gives 19.7 g of white precipitate (Z) with barium chloride solution. (Z) gives carbon-dioxide with an acid.

15. The white solid 'X' is:
(A) metal sulphate
(B) metal chloride
(C) metal carbonate
(D) metal bicarbonate

16. Atomic weight of the metal is:
(A) 23 g/mole
(B) 39.0 g/mole
(C) 6 g/mole
(D) 24 g/mole

Space for Rough work

(Match list Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

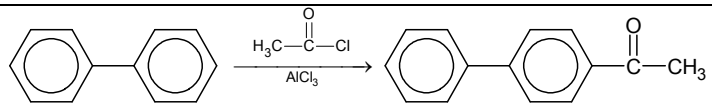
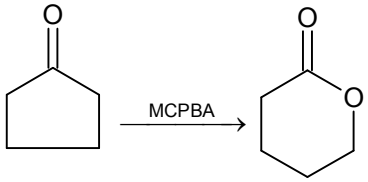
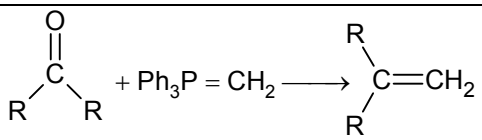
17. Match the List – I with List – II.

List I (Molecule)	**List II (Shape)**
(P) N_2O	(1) Angular
(Q) F_2O	(2) Tetrahedral
(R) BF_4^-	(3) Planar triangular
(S) CH_3^+	(4) Linear

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	3	4
(C)	2	1	4	3
(D)	4	1	2	3

18. Match **List-I** with **List-II**.

List – I Transformation		**List – II Name**	
(P)		(1)	Carbyl amine reaction
(Q)		(2)	Wittin Reaction
(R)		(3)	Bayer-Villiger oxidation
(S)	$Ph-NH_2 \xrightarrow{CHCl_3/KOH} Ph-NC$	(4)	Friedel Craft's Acylation

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	3	4
(C)	2	1	4	3
(D)	4	3	2	1

Space for Rough work

19. Match **List-I** with **List-II**.

List-I		**List-II**	
(P)	$Fe(NO_3)_2$ (aq)	(1)	Only cationic hydrogen
(Q)	$KClO_4$ (aq)	(2)	Only anionic hydrolysis
(R)	HCOONa (aq)	(3)	Both cationic as anionic hydrolysis
(S)	NH_4CN (aq)	(4)	No hydrolysis

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	1	4	2	3
(C)	2	1	4	3
(D)	4	3	2	1

20. Identify the correct matching from the codes given below

Reaction	**Product**
(P) $\text{Borax} \xrightarrow{\Delta}$	(1) $H_2B_4O_7$
(Q) $\text{Borax} + NH_4Cl \xrightarrow{\Delta}$	(2) $NaBO_2 + B_2O_3$
(R) $\text{Borax} \xrightarrow{HCl}$	(3) BN
(S) $\text{Borax} + H_2O$	(4) H_3BO_3

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	3	4
(C)	2	3	4	1
(D)	4	3	2	1

Space for Rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE** than is/are correct

1. Number of ways in which 200 people can be divided in 100 couples is

(A) $\frac{(200)!}{2^{100}(100)!}$ (B) 1.3.5. 199

(C) $\left(\frac{101}{2}\right)\left(\frac{102}{2}\right).....\left(\frac{200}{2}\right)$ (D) $\frac{(200)!}{(100)!}$

2. Consider the quadratic equation $(a + c – b)\,x^2 + 2cx + (b + c – a) = 0$ where a, b, c are distinct real number and $a + c – b \neq 0$. If both the roots of the equation are rational then the numbers which must be rational are

(A) a, b, c (B) $\frac{c}{a-b}$

(C) $\frac{b+c-a}{a+c-b}$ (D) $\frac{b^2}{c-a}$

3. If a, b, c are first three terms of a G.P. if the harmonic mean of a and b is 12 and arithmetic mean of b and c is 3, then

(A) no term of this G.P. is square of an integer (B) arithmetic mean of a, b, c is 3

(C) $b = \pm 6$ (D) common ratio of this G.P. is 2

Space for rough work

4. If the normal at any given point P on ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets its auxiliary circle at Q and R such that $\angle QOR = 90^o$, where O is centre of ellipse, then
(A) $2(a^2 - b^2)^2 = a^4 \sec^2\theta + a^2b^2 \operatorname{cosec}^2\theta$ (B) $a^4 + 5a^2b^2 + 2b^4 = a^4 \tan^2\theta + a^2b^2 \cot^2\theta$
(C) $a^4 + 5b^2a^2 + 2b^4 \geq 2a^3b$ (D) $a^4 + 2b^4 \geq 5a^2b^2 + 2a^3b$

5. If $\begin{vmatrix} x^2 - 5x + 3 & 2x - 5 & 3 \\ 3x^2 + x + 4 & 6x + 1 & 9 \\ 7x^2 - 6x + 9 & 14x - 6 & 21 \end{vmatrix} = ax^3 + bx^2 + cx + d$ then which of the following are correct?
(A) a = 0 (B) b = 0
(C) c = 0 (D) d = 0

6. Let $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + ...$ If A_0, A_1, A_2 are in A.P. then the value of n is
(A) 2 (B) 3
(C) 5 (D) 7

7. A line passes through the points whose position vectors are $\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} + \hat{k}$. The position vector of a point on it at a unit distance from the first point is/are
(A) $\frac{1}{5}(5\hat{i} + \hat{j} - 7\hat{k})$ (B) $\frac{1}{5}(5\hat{i} + 9\hat{j} - 13\hat{k})$
(C) $\frac{1}{5}(6\hat{i} + \hat{j} - 7\hat{k})$ (D) $\frac{1}{5}(4\hat{i} + 9\hat{j} - 13\hat{k})$

8. Let $u = \int_0^\infty \frac{dx}{x^4 + 7x^2 + 1}$ and $v = \int_0^\infty \frac{x^2\, dx}{x^4 + 7x^2 + 1}$ then
(A) v > u (B) $6v = \pi$
(C) $3u + 2v = 5\pi/6$ (D) $u + v = \pi/3$

Space for rough work

Comprehension Type

This section contains **4 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 9 to 10

Read the following write up carefully and answer the following questions:
Continuous probability distributions, a continuous distribution is one in which the variate may take any value between certain limits a and b, a < b. Suppose that the probability of the variate X falling in the infinitesimal interval $x-\frac{1}{2}dx$ to $x+\frac{1}{2}dx$ is expressible as f(x) dx, where f(x) is a continuous function of x

Symbolically, $P(x-\frac{1}{2}dx \le X \le x+\frac{1}{2}dx) = f(x)\,dx$ where f(x) is called the probability density function (abbreviated as p.d.f.) or simply density function. The continuous curve y = f(x) is called probability curve; and when this is symmetrical, the distribution is said to be symmetrical. Clearly, the probability density function possesses the following properties

(i) $f(x) \ge 0$ for every x in the interval [a, b], a < b

(ii) $\int_a^b f(x)\,dx = 1$, a, b > 0 since the total area under the curve is unity

(iii) Furthermore, we define for any [c, d], where c, d $\in$ [a, b], c < d;

$P(c \le X \le d) = \int_c^d f(x)\,dx$ ..... (1)

We define F(x), the cumulative distribution function (abbreviated as c.d.f.) of the random variate X where $F(x) = P(X \le x)$

or $F(x) = \int_a^x f(x)\,dx$ ..... (2)

9. If $f(x) = \begin{cases} 2x & ; \ 0 \le x \le 1 \\ 0 & ; \ x > 1 \end{cases}$ then the probability that $x \le \frac{1}{2}$ is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$

(C) $\frac{3}{4}$ (D) $\frac{1}{8}$

10. Suppose the life in hours (x) of a certain kind of radio tube has the probability density function $f(x) = \frac{100}{x^2}$ when x > 100 and zero when x < 100. Then the probability that none of three such tubes in a given radio set will have to be replaced during the first 150 hours of operation, is

(A) $\frac{1}{27}$ (B) $\frac{8}{27}$

(C) $\frac{1}{225}$ (D) $\frac{26}{27}$

Space for rough work

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:
Let f be a function defined so that every element of the co-domain has at most two pre-images and there is at least one element in the co-domain which has exactly two pre-images we shall call this function as "two-one" function. A two-one function is definitely a many one function but vice-versa is not true. For example, $y = |e^x - 1|$ is a "two-one" function. $y = x^3 - x$ is a many one function but not a "two-one" function. In the light of above definition answers the following questions:

11. In the following functions which one is a "two-one" function
(A) $y = \left|\ell n|x|\right|$ (B) $y = x^2 \sin x$
(C) $y = x^3 + 3x + 1$ (D) $y = x^4 - x + 1$

12. Let f(x) = {x} be the fractional part function. For what domain is the function "two-one"?
(A) $\left[\frac{1}{2}, \frac{5}{2}\right]$ (B) $\left[-\frac{1}{2}, \frac{3}{2}\right)$
(C) [1, 2) (D) none of these

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:
If $y = \int_{u(x)}^{v(x)} f(t)\,dt$, let us define $\frac{dy}{dx}$ in a different manner as $\frac{dy}{dx} = v'(x)\, f^2(v(x)) - u'(x)\, f^2(u(x))$ and the equation of the tangent at (a, b) as $y - b = \left(\frac{dy}{dx}\right)_{(a,b)} (x - a)$

13. If $y = \int_{x}^{x^2} t^2\,dt$, then equation of tangent at x = 1 is
(A) y = x + 1 (B) x + y = 1
(C) y = x – 1 (D) y = x

14. If $F(x) = \int_{1}^{x} e^{t^2/2}(1 - t^2)\,dt$, then $\frac{d}{dx}F(x)$ at x = 1 is
(A) 0 (B) –1
(C) 1 (D) 2

Space for rough work

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:

In reducing a given equation to the standard forms $(\sin x = \sin\alpha$ etc$)$ we apply several trigonometric or algebraic transformations. As a result of which the canonical form finally obtained may not be equivalent to the original equation resulting either in loss of solutions or in the appearance of fake solutions

15. The solution set of the equation $\sqrt{5-2\sin x} = 6\sin x - 1$ is

(A) $n\pi + (-1)^n \sin^{-1}\left(-\frac{2}{9}\right)$ or $x = n\pi + (-1)^n \frac{\pi}{6}$ (B) $n\pi + (-1)^n \frac{\pi}{6}$

(C) $n\pi + (-1)^n \sin^{-1}\left(\frac{2}{9}\right)$ (D) Null set

16. The equation $2\cot 2x - 3\cot 3x = \tan 2x$ has

(A) two solutions in $\left(0, \frac{\pi}{3}\right)$ (B) one solution in $\left(0, \frac{\pi}{3}\right)$

(C) no solution in $(-\infty, \infty)$ (D) none of these

(Match list Type)

This section contains **4 multiple choice questions.** Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

17. Matching the following List-I with List-II.

	List – I	List – II
(P)	The least positive integral solution of $x^2 - 4x > \cot^{-1} x$	1. 1
(Q)	The least positive integral value of a for which $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$ is increasing	2. 2
(R)	Let f: R $\to$ R be such that f(a) = 1, f′(a) = 2, then $\lim_{x \to 0} \left(\frac{f^2(a+x)}{f(a)}\right)^{\frac{1}{x}} = e^k$, then k is equal to	3. 5
(S)	Number of integral values of x which satisfy equation $\sin^{-1}((3 - x)(x - 1)) + \sin^{-1}(2 - \lvert x\rvert) = \pi/2$ is/are	4. 4

Codes:

	P	Q	R	S
(A)	3	1	2	4
(B)	1	2	3	4
(C)	4	3	2	1
(D)	3	1	4	2

Space for rough work

18. Matching the following List-I with List-II.

	List – I	List – II
(P)	If $y = 2x + c - 4$ is a normal to the parabola $y^2 = 4x$, then value of '–c' is	1. 12
(Q)	If the mid point of a chord of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ is (0, 3), then length of the chord is $\frac{4k}{5}$, then k + 4 is equal to	2. 8
(R)	The product of the lengths of the perpendiculars from the two focii on any tangent to the hyperbola $\frac{x^2}{25} - \frac{y^2}{3} = 1$ is $\sqrt{k}$, then k is	3. 3
(S)	The line $x + y = a$ touches the hyperbola $x^2 - 2y^2 = 18$, if a is equal to ±b, then value of |b| is	4. 9

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	3	4
(C)	2	1	4	3
(D)	4	3	2	1

19. Matching the following List-I with List-II.

	List – I	List – II
(P)	If the sum of unit vectors is a unit vector, then magnitude of difference is $\sqrt{x}$, then the value of x is	1. $\frac{1}{2}$
(Q)	If $f(x) = \sin x$, $\forall\ x \in \left[0, \frac{\pi}{2}\right]$, $f(x) + f(\pi - x) = 2$, $\forall\ x \in \left(\frac{\pi}{2}, \pi\right]$ and $f(x) = f(2\pi - x)$, $\forall\ x \in (\pi, 2\pi)$, then the area enclosed by $y = f(x)$ and the x-axis is $k\pi$ sq. units. Then k is	2. 2
(R)	Let f and g be differentiable function satisfying $g'(a) = 2$, $g(a) = b$ and $g = f^{-1}$ then $f'(b)$ is equal to	3. 3
(S)	The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ w.r.t $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at x = 0 is	4. $\frac{1}{4}$

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	3	2	1	4
(C)	3	1	4	2
(D)	4	3	2	1

Space for rough work

20. Matching the following List-I with List-II

List – I		List – II	
(P)	If a, b, c $\in$ N and a, b, c $\leq$ 100, then probability that a, b, c are individually divisible by both 2 and 3	1.	$\frac{5}{28}$
(Q)	Probability that in a year of 22nd century chosen at random there will be 53 Sundays	2.	$\frac{31}{32}$
(R)	If a unbiased coin is tossed 5 times then probability of getting at least 1 head is	3.	$\frac{5}{12}$
(S)	If $P(A \cup B) = \frac{3}{4}, P(A \cap B) = \frac{1}{4}$, $P(\bar{A}) = \frac{2}{3}$, then $P(\bar{A} \cap B)$ is	4.	$\frac{4}{1155}$

Codes:

	P	Q	R	S
(A)	4	1	3	2
(B)	1	2	3	4
(C)	3	4	2	1
(D)	4	1	2	3

Space for rough work